Subject-Level Calibration Heterogeneity is Stable Across LLM Families and Scales: A Statistical Audit of MMLU

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Abstract

Aggregate calibration metrics obscure a critical risk: LLMs are most overconfident in the specialized domains—legal reasoning, medical science, technical STEM—where users are least equipped to catch errors independently. We develop a statistical audit framework—a multilevel decomposition of question-level calibration gap variance, isotonic reliability curves per subject, empirical Bayes shrinkage, and Benjamini–Hochberg FDR correction—and apply it to MMLU across seven instruction-tuned models from two families spanning six parameter scales: GPT-4o-mini, Llama-3-8B-Instruct, and Qwen-2.5-{0.5B, 1.5B, 3B, 7B, 14B}-Instruct. Three findings emerge. (i) Among the six larger models, subject-level ECE rankings are highly correlated across all 15 model pairs (Spearman $\rho \in [0.78, 0.94]$), with cross-family agreement at matched capability matching the strongest within-family pairs. (ii) The same subjects are robustly worst-calibrated (college-level quantitative STEM, niche factual recall, structured ethical/legal reasoning) and robustly best-calibrated (high-school-level Social Sciences and applied/factual subjects), revealing specialization level as a meaningful axis of calibration quality. (iii) Within the Qwen family at matched precision, the entropy coefficient on the calibration gap rises monotonically across the 1.5B$\rightarrow$3B$\rightarrow$7B$\rightarrow$14B scaling sweep, with confidence growing faster than accuracy as items become harder—qualitatively consistent with prior reports of RLHF-induced confidence saturation (Kadavath et al., 2022; Nakkiran et al., 2025). The smallest model in our set (Qwen-0.5B) shows extreme positional bias and a qualitatively different miscalibration pattern; we read it as a sign that our elicitation protocol does not extend cleanly below approximately 1B parameters. The stability of these patterns across the larger model space provides empirical motivation for subject-aware recalibration as in Luo et al. (2025), and the audit itself is a transferable diagnostic for new models.

1. Introduction

When users consult an LLM for legal guidance, medical information, or technical scientific questions, they typically lack the domain expertise to evaluate the answer independently—placing full weight on the model's expressed confidence as a reliability signal. Systematic overconfidence in these domains therefore creates a predictable, population-scale harm: users acting on high-confidence wrong answers have no reliable way to detect the error. Our findings make this concrete: across seven models from two families spanning a $28\times$ scale range, the most severely miscalibrated subjects are stably the professional and technical domains—`professional_law`, `virology`, `college_chemistry`, `college_mathematics`—precisely where reliable confidence signals matter most. Yet the standard calibration literature reports only aggregate scalars—ECE (Guo et al., 2017), NLL, or related—that are silent about where miscalibration concentrates. We ask three questions:

1. **Where is calibration good and bad within a benchmark?**
2. **Are these patterns stable across models?**
3. **What mechanisms produce the observed heterogeneity?**

Our contribution is a statistical audit framework—multilevel decomposition, FDR correction, isotonic curves with empirical Bayes shrinkage—that produces direct answers to these questions, and an application of the framework to seven instruction-tuned LLMs on MMLU (Hendrycks et al., 2021). The framework is diagnostic rather than corrective: Luo et al. (2025) demonstrates that subject-aware temperature scaling improves on global recalibration; we ask whether

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the structural assumption underlying that scoping decision (that miscalibration patterns are stable enough to be worth scoping over) is empirically justified.

Prior work has studied LLM calibration mostly at the aggregate level. Kadavath et al. (2022) showed that token-level logprobs carry a meaningful calibration signal on multiple-choice tasks and that instruction tuning tends to degrade it; Xiong et al. (2024) noted calibration degrades on tasks requiring professional knowledge but did not investigate subject-level structure formally. Nakkiran et al. (2025) and Yaldiz et al. (2026) document RLHF-induced confidence saturation at the aggregate level. We bring a formal statistical audit to the within-benchmark heterogeneity these works describe only in passing.

2. Method

Confidence elicitation. For each MMLU question we extract token-level log probabilities over $\{A, B, C, D\}$, softmax-renormalize, and define c_i as the maximum renormalized probability. The signed calibration gap $\mathrm{gap}_i = c_i - y_i$ (where $y_i \in \{0, 1\}$ is correctness) is our primary outcome. GPT-4o-mini is queried via the OpenAI Batch API with `top_logprobs=20`; the open-weight models are run on a Colab GPU with the assistant turn pre-filled by "The answer is" to score the next-token position. The 3B, 7B, 8B, and 14B models use 4-bit quantization (BitsAndBytes; Dettmers et al. 2022); the 1.5B model is run twice (4-bit and fp16) so the precision effect can be examined directly, with the fp16 version used as the canonical 1.5B in the main analysis. Test-set accuracies: 0.749, 0.606, 0.405, 0.571, 0.601, 0.685, 0.757 for GPT, Llama, Qwen-{0.5B, 1.5B fp16, 3B, 7B, 14B}. Qwen-0.5B exhibits extreme positional bias (predicting D 59% of the time vs. A 5.6%), a confound we discuss where it affects results.

Multilevel decomposition. We fit a three-level mixed-effects model with questions nested in subjects nested in domains:

$$\mathrm{gap}_{ijk} = \mu + \mathbf{x}_{ijk}^\top \boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk}, \qquad (1)$$

where u_k is a domain random intercept, v_{jk} a subject-within-domain intercept, and $\mathbf{x}_{ijk}$ standardized question-level covariates: word count, max option length, negation indicator, and Shannon entropy of the A/B/C/D distribution ($H_i = -\sum_k p_{ik} \log p_{ik}$ in nats). Variance components are estimated by REML; the ICC decomposition reports the share of total variance at each level.

Calibration curves and ECE. For each subject we fit isotonic regression (Zadrozny & Elkan, 2002) and an unconstrained Gaussian kernel smoother ($\sigma = 1.5$ bins) on (c_i, y_i), and compute ECE as the 20-bin weighted absolute deviation between the curve and the diagonal.

Multiple testing. For each of 57 subjects we test H_0: mean calibration gap $= 0$ via a one-sample t-test on question-level gaps, then apply Benjamini–Hochberg FDR correction at $\alpha = 0.05$.

3. Results

3.1. Variance decomposition (RQ1)

Table 1 reports the ICC decomposition. Across all seven models the bulk of calibration-gap variance lies at the question level (92.1–98.3%) and domain-level variance collapses to zero. Between-subject ICCs span nearly a factor of five, from 1.7% (Qwen-1.5B) to 7.9% (GPT-4o-mini). Within the Qwen family at matched precision (1.5B fp16, 3B 4-bit, 7B 4-bit, 14B 4-bit), ICC jumps from 1.5B to 3B and is then roughly flat: $0.017 \rightarrow 0.051 \rightarrow 0.049 \rightarrow 0.059$.

Table 1. Variance decomposition of the calibration gap. Domain-level variance is estimated at zero for every model; $\mathrm{ICC}_{\text{subject}}$ is the share of total variance at the subject level.

Model	$\sigma^2_{\text{subject}}$	σ^2_{ε}	$\mathrm{ICC}_{\text{subject}}$
GPT-4o-mini	0.013	0.150	0.079
Llama-3-8B-Instruct	0.004	0.190	0.020
Qwen-2.5-0.5B-Instruct	0.004	0.224	0.018
Qwen-2.5-1.5B-Instruct	0.003	0.197	0.017
Qwen-2.5-3B-Instruct	0.011	0.203	0.051
Qwen-2.5-7B-Instruct	0.009	0.177	0.049
Qwen-2.5-14B-Instruct	0.010	0.156	0.059

3.2. Where calibration succeeds vs. fails (RQ1)

The same subject clusters appear at both extremes of the ECE distribution across the six larger models (Qwen-0.5B is excluded from the cluster analysis due to its positional-bias confound; Section 3.3). Four subjects are top-10 most miscalibrated for every model: `virology`, `college_chemistry`, `college_physics`, `professional_law`. Top-15 expansion adds `college_mathematics`, `econometrics`, `high_school_physics`, `machine_learning`, and `moral_scenarios`. These cluster into three semantic groups: college-level quantitative STEM, niche factual recall, and structured ethical/legal reasoning—precisely the high-stakes domains where systematic overconfidence creates the largest harm potential.

Symmetrically, two subjects are bottom-5 best-calibrated for every larger model (`high_school_government_and_politics`, `high_school_psychology`); bottom-10 adds `high_school_us_history`, `marketing`, and `miscellaneous`. Two patterns:

- **Specialization level matters.** MMLU's high-school-

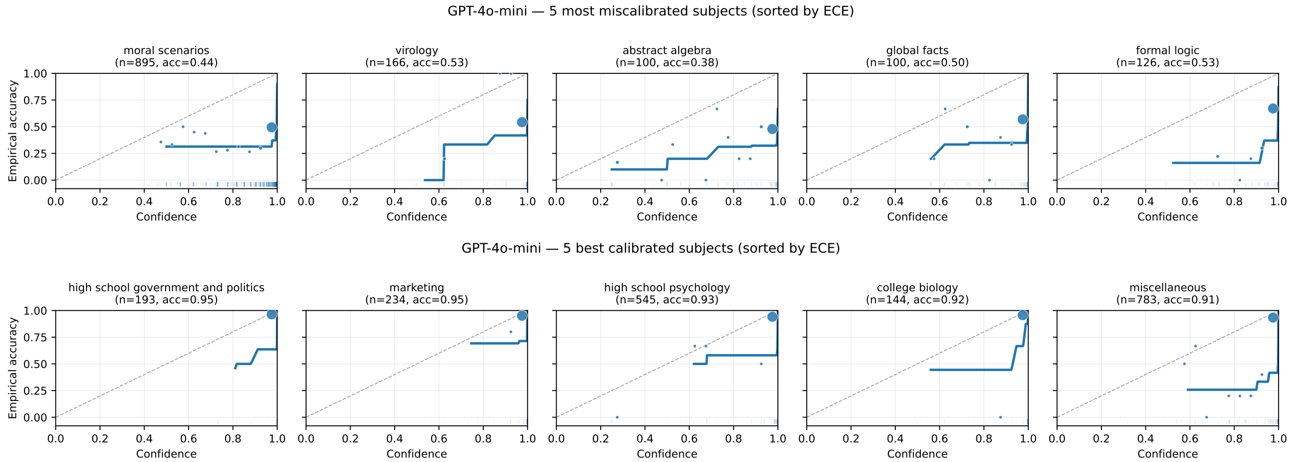

Figure 1. GPT-4o-mini reliability curves. **Top**: five most miscalibrated subjects by ECE. **Bottom**: five best-calibrated subjects. Same conventions in both: solid line is isotonic regression, plotted only on observed support; markers are 20-bin empirical accuracy sized by bin count; dashed line is the calibration diagonal.

level subjects systematically show smaller calibration gaps than their college-level counterparts (compare `high_school_psychology` to professional and college subjects in similar areas).

- **Domain category matters.** Best-calibrated subjects are dominated by Social Sciences and the "Other" (applied/factual) MMLU category; no STEM subject appears in the bottom-10 intersection across models.

Figure 1 contrasts top-5 most-miscalibrated and bottom-5 best-calibrated reliability curves for GPT-4o-mini. Curves are masked to the 1st–99th percentile of observed confidence per subject to avoid displaying isotonic extrapolation in regions with no data. The two panels reveal the heterogeneity that aggregate ECE conceals: in a single model on a single benchmark, some subjects sit on the calibration diagonal while others fall sharply below it.

3.3. Cross-model agreement (RQ2)

Figure 2 reports the 7×7 Spearman rank correlation matrix of subject-level ECE. Among the six larger models, all 15 pairwise correlations fall in $[0.78, 0.94]$, with the strongest pair Qwen-7B vs Qwen-14B at 0.94 and the cross-family GPT-4o-mini vs Qwen-14B and GPT-4o-mini vs Qwen-7B both at 0.92. Subject-level miscalibration patterns are therefore consistent across model families and scales above approximately 1B parameters.

Qwen-0.5B is the conspicuous exception: its correlations with the other six models fall in $[0.16, 0.22]$, with bootstrap 95% CIs that include zero. The underlying cause is extreme positional bias—Qwen-0.5B predicts D on 59% of questions and A on only 5.6%, vs. approximately uniform 25% for the larger models. The 0.5B's miscalibration is therefore dominated by a spurious preference for the D position rather than genuine knowledge gaps, producing a qualitatively distinct ranking of which subjects appear miscalibrated. We interpret this as evidence that, below approximately 1B parameters, positional bias dominates the calibration signal in our elicitation protocol.

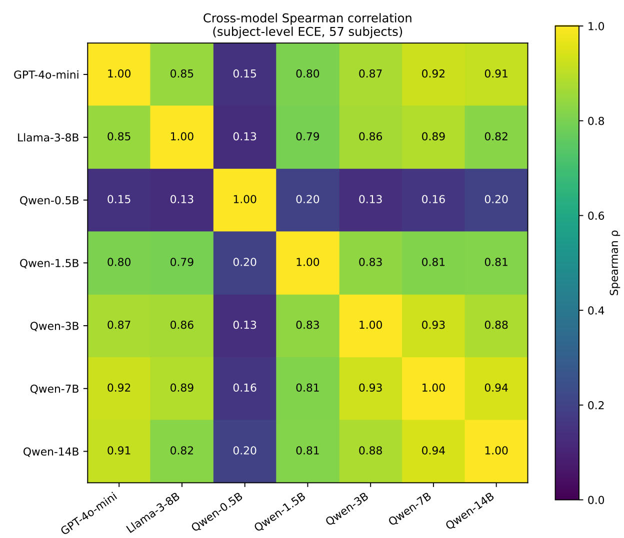

Figure 2. Spearman rank correlation matrix of subject-level ECE across the seven models (57 common subjects). Among the six larger models, all 15 pairwise correlations fall in $[0.78, 0.94]$. Qwen-0.5B's correlations with the others are $[0.16, 0.22]$, reflecting a qualitatively different miscalibration pattern driven by extreme positional bias.

3.4. Mechanism: predictors and within-family scale (RQ3)

Table 2 reports the fixed effects from the multilevel model. Intercepts confirm large mean overconfidence in every model (0.179–0.311). Within the Qwen family the intercept is non-monotonic in scale, peaking at 3B (0.311) and declining at 7B (0.233) and 14B (0.210); we do not read this as a clean scaling trend on its own.

Table 2. Fixed effects from the three-level mixed model. Continuous covariates standardized. Significance: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.10$.

Model	Intercept	Word ct	Max opt	Negation	Entropy
GPT-4o-mini	0.196***	0.031***	0.001	-0.001	0.016***
Llama-3-8B	0.197***	-0.005	-0.001	0.007	0.008†
Qwen-0.5B	0.209***	-0.013*	-0.028***	0.016	-0.063***
Qwen-1.5B	0.179***	0.003	0.005	-0.010	-0.004
Qwen-3B	0.311***	0.010	-0.008†	-0.014	0.006
Qwen-7B	0.233***	0.019**	-0.007	-0.001	0.029***
Qwen-14B	0.210***	0.023***	-0.007	0.005	0.040***

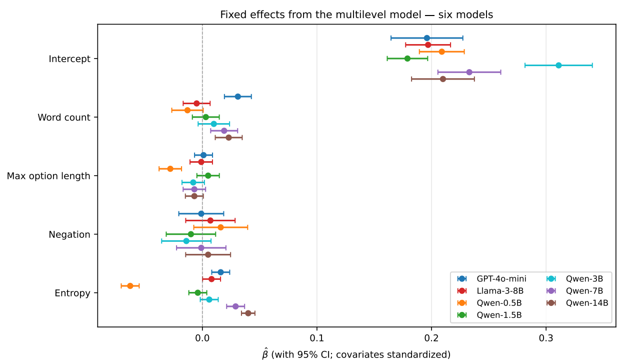

Figure 3. Fixed effect estimates with 95% confidence intervals across the seven models. The entropy coefficient progression—from strongly negative at Qwen-0.5B through near-zero at Qwen-1.5B to strongly positive at the larger Qwens—is the most informative pattern.

Entropy: confidence outpaces accuracy on hard questions at scale. Entropy is the most informative predictor. Within the Qwen family at matched precision its coefficient on the calibration gap rises with scale: -0.004 (n.s.) at 1.5B, $+0.006$ (n.s.) at 3B, $+0.029$ at 7B, $+0.040$ at 14B (last two $p < 0.001$). Figure 4 traces the mechanism by partitioning each model's questions into entropy quartiles and plotting mean confidence (solid) and mean accuracy (dashed). At 1.5B the two curves move roughly together across quartiles, so the gap is nearly flat in entropy (Q1 vs. Q4: $+0.09$ vs. $+0.13$). At 7B and 14B confidence stays near saturation across all four quartiles while accuracy still drops with entropy as expected, so the gap widens on the model's harder items (Q1 vs. Q4: $+0.05$ vs. $+0.37$ at 7B; $+0.03$ vs. $+0.46$ at 14B). The qualitative pattern is one of confidence growing faster than accuracy as both model scale and item entropy increase, rather than accuracy itself collapsing on hard items.

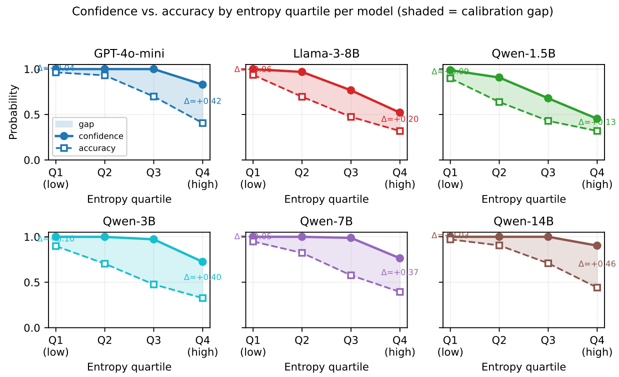

Figure 4. Confidence (solid) and accuracy (dashed) by entropy quartile for the six larger models (Qwen-0.5B omitted; see Section 3.3). Shaded area is the calibration gap. Within Qwen, the curves move roughly together at 1.5B, but separate progressively at 7B and 14B as confidence stays near saturation across quartiles while accuracy still falls with entropy.

This is the qualitative pattern reported at the aggregate level for RLHF-tuned models (Kadavath et al., 2022; Nakkiran et al., 2025; Yaldiz et al., 2026). Two caveats limit how much weight we put on the within-family comparison: it rests on four observational checkpoints, and Qwen-0.5B's strongly negative entropy coefficient (-0.063, $p < 0.001$) is best read as an artifact of its positional bias rather than a fourth scale point. Broader claims would require additional small or base-model checkpoints.

Figure 5 summarizes the within-Qwen scaling on the four matched-precision checkpoints (1.5B fp16, 3B 4-bit, 7B 4-bit, 14B 4-bit). The entropy coefficient rises monotonically; the between-subject ICC jumps from 1.5B to 3B and is essentially flat thereafter; the mean-overconfidence intercept is non-monotonic.

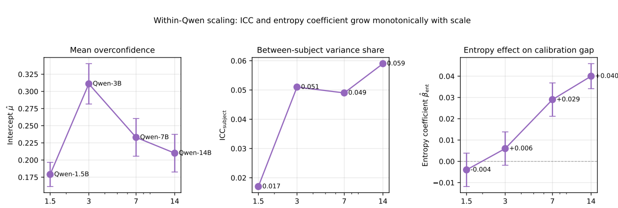

Figure 5. Within-Qwen scaling on the four matched-precision checkpoints. The entropy coefficient rises monotonically with scale; ICC jumps from 1.5B to 3B and then plateaus; the intercept is non-monotonic.

3.5. FDR-corrected rankings and sensitivity

At $\alpha = 0.05$: GPT-4o-mini, Llama, Qwen-3B, Qwen-7B, and Qwen-14B reject all 57 subjects; Qwen-1.5B (fp16) rejects 56/57, and Qwen-0.5B rejects 55/57. The substantive finding is the magnitude of heterogeneity: per-model gap ranges span 5.4–13.0× from the best-calibrated to the worst-calibrated subject in the larger six models.

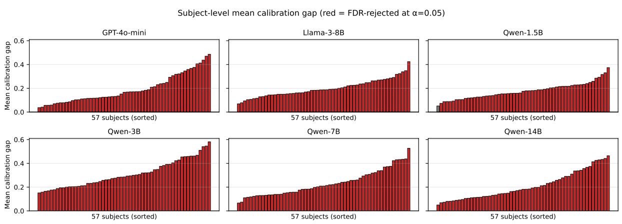

Figure 6. Mean calibration gap per subject, sorted ascending, for the six larger models (Qwen-0.5B omitted; see Section 3.3). Red bars are FDR-rejected at $\alpha = 0.05$; gray bars are not. Five of six models reject all 57 subjects; Qwen-1.5B (fp16) rejects 56/57.

Sensitivity-analysis conclusions extend uniformly: ECE rankings are stable across bin counts (Spearman ≥ 0.97 vs. 20-bin baseline), proper-scoring NLCS rankings track ECE rankings (Spearman ≥ 0.79), and two-level vs. three-level ICC estimates are numerically identical (consistent with $\sigma^2_{\text{domain}} = 0$).

4. Discussion

What the audit recovers. Aggregate ECE conceals structured heterogeneity that is statistically real (FDR-rejected, robust to bin count, robust to proper-scoring metrics), task-driven (correlated across families and scales above 1B parameters), and aligned with intuitive content axes (specialization level, domain category). A consistent best-calibrated set (high-school Social Sciences and applied subjects) and worst-calibrated set (college-level quantitative STEM, niche factual recall, structured reasoning) appear in every model above 1B parameters in our sample. Within the Qwen family the entropy coefficient grows monotonically with scale, while the between-subject ICC plateaus from 3B onward.

Implications for recalibration. The cross-model stability we observe is the kind of evidence one would want before scoping a correction at the subject level rather than globally. We do not re-fit corrections here: Luo et al. (2025) demonstrates that subject-aware temperature scaling improves on global recalibration, and our findings provide an upstream diagnostic for when that scoping is empirically motivated. The robustly worst-calibrated subjects can also serve as a small targeted set for flagging miscalibration on a new model before deployment.

The audit as an accountability instrument. Because the worst-calibrated subjects are stable across families and scales in our data, the protocol is portable: a deployer or regulator can re-run it on a new model and read off subject-level overconfidence directly, without re-establishing baselines. The recurring offenders in our sample—`professional_law`, `virology`, `college_chemistry`, `moral_scenarios`—are a natural starting point for pre-deployment review in legal, medical, and other high-stakes contexts where aggregate metrics like top-line ECE would mask the structured risk.

Limitations. The within-family scale comparison rests on four observational checkpoints (Qwen 1.5B, 3B, 7B, 14B); additional scale points and base-model checkpoints would be needed to make a stronger claim. Cross-family generalization would benefit from a third family (e.g., Mistral-7B or Gemma-2-9B) at matched scale. 4-bit quantization affects five of the seven models; we ran a precision comparison at 1.5B (4-bit vs. fp16) but not at larger scales. Qwen-0.5B's extreme positional bias prevents using it as a fifth scaling point. All findings are specific to MMLU's four-choice format and do not directly extend to open-ended generation.

5. Conclusion

Subject-level calibration heterogeneity in LLMs is large and stable across model families above approximately 1B parameters in our sample. Applied/factual high-school subjects are reliably well-calibrated; college-level technical content, niche factual recall, and structured ethical/legal reasoning are reliably poorly calibrated. Subject-level ECE rankings correlate strongly across all 15 pairs of larger models ($\rho \in [0.78, 0.94]$), including cross-family pairs at matched capability, pointing to task structure rather than any particular training pipeline. Within the Qwen family the entropy coefficient grows monotonically with scale, with confidence outpacing accuracy on harder questions—qualitatively in line with prior reports of RLHF-induced confidence saturation. The audit provides empirical motivation for subject-aware recalibration scoping, a portable pre-deployment diagnostic, and a concrete accountability instrument for high-stakes deployments where calibration failures carry the greatest societal cost.

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